

# Demo

May 8, 2026

```
[34]: import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
[7]: df = pd.read_csv('mail_data.csv')
```

```
[8]: print(df)
```

|      | Category | Message                                           |
|------|----------|---------------------------------------------------|
| 0    | ham      | Go until jurong point, crazy.. Available only ... |
| 1    | ham      | Ok lar... Joking wif u oni...                     |
| 2    | spam     | Free entry in 2 a wkly comp to win FA Cup fina... |
| 3    | ham      | U dun say so early hor... U c already then say... |
| 4    | ham      | Nah I don't think he goes to usf, he lives aro... |
| ...  | ...      | ...                                               |
| 5567 | spam     | This is the 2nd time we have tried 2 contact u... |
| 5568 | ham      | Will ü b going to esplanade fr home?              |
| 5569 | ham      | Pity, * was in mood for that. So...any other s... |
| 5570 | ham      | The guy did some bitching but I acted like i'd... |
| 5571 | ham      | Rofl. Its true to its name                        |

[5572 rows x 2 columns]

```
[9]: data = df.where((pd.notnull(df)), '')
```

```
[10]: data.head()
```

```
[10]: Category      Message
0      ham  Go until jurong point, crazy.. Available only ...
1      ham  Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3      ham  U dun say so early hor... U c already then say...
4      ham  Nah I don't think he goes to usf, he lives aro...
```

```
[11]: data.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Category    5572 non-null   object
1   Message     5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB

```

```
[12]: data.shape
```

```
[12]: (5572, 2)
```

```
[18]: print(data.columns)
```

```
Index(['Category', 'Message'], dtype='object')
```

```
[19]: data.loc[data['Category'] == 'spam', 'Category',] = 0
data.loc[data['Category'] == 'ham', 'Category',] = 1
```

```
[20]: X=data['Message']
```

```
Y=data['Category']
```

```
[21]: print(X)
```

```

0      Go until jurong point, crazy.. Available only ...
1                Ok lar... Joking wif u oni...
2      Free entry in 2 a wkly comp to win FA Cup fina...
3      U dun say so early hor... U c already then say...
4      Nah I don't think he goes to usf, he lives aro...
...
5567    This is the 2nd time we have tried 2 contact u...
5568                Will ü b going to esplanade fr home?
5569    Pity, * was in mood for that. So...any other s...
5570    The guy did some bitching but I acted like i'd...
5571                Rofl. Its true to its name
Name: Message, Length: 5572, dtype: object

```

```
[22]: print(Y)
```

```

0      1
1      1
2      0
3      1
4      1
..

```

```

5567    0
5568    1
5569    1
5570    1
5571    1
Name: Category, Length: 5572, dtype: object

```

```
[23]: X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state = 3)
```

```
[24]: print(X.shape)
      print(X_train.shape)
      print(X_test.shape)
```

```

(5572,)
(4457,)
(1115,)

```

```
[25]: print(Y.shape)
      print(Y_train.shape)
      print(Y_test.shape)
```

```

(5572,)
(4457,)
(1115,)

```

```
[26]: feature_extraction = TfidfVectorizer(min_df=1, stop_words='english', lowercase=True)
```

```

X_train_features = feature_extraction.fit_transform(X_train)
X_test_features = feature_extraction.transform(X_test)
Y_train = Y_train.astype('int')
Y_test = Y_test.astype('int')

```

```
[27]: print(X_train)
```

```

3075          Don know. I didn't msg him recently.
1787    Do you know why god created gap between your f...
1614          Thnx dude. u guys out 2nite?
4304          Yup i'm free...
3266    44 7732584351, Do you want a New Nokia 3510i c...
...
789      5 Free Top Polyphonic Tones call 087018728737,...
968      What do u want when i come back?.a beautiful n...
1667     Guess who spent all last night phasing in and ...
3321     Eh sorry leh... I din c ur msg. Not sad ahead...
1688     Free Top ringtone -sub to weekly ringtone-get ...
Name: Message, Length: 4457, dtype: object

```

```
[28]: print(X_train_features)
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'  
  with 34775 stored elements and shape (4457, 7431)>
```

| Coords       | Values              |
|--------------|---------------------|
| (0, 2329)    | 0.38783870336935383 |
| (0, 3811)    | 0.34780165336891333 |
| (0, 2224)    | 0.4131033779433779  |
| (0, 4456)    | 0.4168658090846482  |
| (0, 5413)    | 0.6198254967574347  |
| (1, 3811)    | 0.17419952275504036 |
| (1, 3046)    | 0.25037127926135183 |
| (1, 1991)    | 0.33036995955537024 |
| (1, 2956)    | 0.33036995955537024 |
| (1, 2758)    | 0.32264078859437995 |
| (1, 1839)    | 0.2784903590561455  |
| (1, 918)     | 0.22871581159877652 |
| (1, 2746)    | 0.33982970028640835 |
| (1, 2957)    | 0.33982970028640835 |
| (1, 3325)    | 0.31610586766078863 |
| (1, 3185)    | 0.2969448295769459  |
| (1, 4080)    | 0.18880584110891166 |
| (2, 6601)    | 0.6056811524587516  |
| (2, 2404)    | 0.45287711070606745 |
| (2, 3156)    | 0.4107239318312698  |
| (2, 407)     | 0.509272536051008   |
| (3, 7414)    | 0.8100020912469564  |
| (3, 2870)    | 0.5864269879324768  |
| (4, 2870)    | 0.41872147309323754 |
| (4, 487)     | 0.2899118421746198  |
| :            | :                   |
| (4454, 2855) | 0.472106650836418   |
| (4454, 2246) | 0.472106650836418   |
| (4455, 4456) | 0.24920025316220423 |
| (4455, 3922) | 0.31287563163368587 |
| (4455, 6916) | 0.19636985317119715 |
| (4455, 4715) | 0.30714144758811196 |
| (4455, 3872) | 0.3108911491788658  |
| (4455, 7113) | 0.30536590342067704 |
| (4455, 6091) | 0.23103841516927642 |
| (4455, 6810) | 0.29731757715898277 |
| (4455, 5646) | 0.33545678464631296 |
| (4455, 2469) | 0.3544154551183794  |
| (4455, 2247) | 0.37052851863170466 |
| (4456, 2870) | 0.3152319627311339  |
| (4456, 5778) | 0.16243064490100795 |
| (4456, 334)  | 0.22200777116549383 |

```
(4456, 6307) 0.2752760476857975
(4456, 6249) 0.17573831794959718
(4456, 7150) 0.36775546814476695
(4456, 7154) 0.24083218452280056
(4456, 6028) 0.21034888000987118
(4456, 5569) 0.46193954042991725
(4456, 6311) 0.3013318243170762
(4456, 647) 0.3013318243170762
(4456, 141) 0.292943737785358
```

```
[29]: model = LogisticRegression()

model.fit(X_train_features, Y_train)
```

```
[29]: LogisticRegression()
```

```
[30]: prediction_on_training_data = model.predict(X_train_features)
accuracy_on_training_data = accuracy_score(Y_train, prediction_on_training_data)

print('Accuracy on training data : ', accuracy_on_training_data)
```

```
Accuracy on training data : 0.9676912721561588
```

```
[31]: prediction_on_test_data = model.predict(X_test_features)
accuracy_on_test_data = accuracy_score(Y_test, prediction_on_test_data)

print('Accuracy on test data : ', accuracy_on_test_data)
```

```
Accuracy on test data : 0.9668161434977578
```

```
[42]: input_your_mail = ["Even my brother is not like to speak with me. They treat me_
↳like aids patent."]

input_data_features = feature_extraction.transform(input_your_mail)

prediction = model.predict(input_data_features)

print(prediction)
if prediction[0] == 1:
    print("Ham Mail")
else:
    print("Spam Mail")
```

```
[1]
Ham Mail
```

```
[ ]:
```

[ ]: